+1 438-410-6013 ; +33 7 81 34 62 90
Montreal / Paris (open to relocation)
elouan.salaun1909@gmail.com

Elouan Salaun

Data Scientist / Engineer

Portfolio/Blog
LinkedIn
GitHub

EXPERIENCE

Data Scientist – Research Partner | Commissionnaires du Québec — Montreal, Sept. 2024 – Juil. 2026

- Designed and delivered an **end-to-end AI-driven tool** covering the entire bid response process; presented it several times to the executive committee and gave an AI talk (technologies, integration, risks…) to all employees.
- Developed a pricing model (XGBoost / Learning-to-Rank) combining market dynamics with the technical characteristics of each contract: simulated win rate of **40% vs. 15%** for the existing pricing rule, **+10% profit** compared with historical results in retrospective backtests.
- Built the **data pipelines** from scratch (internal Excel and PDF files, external APIs and web scraping) and the cross-referenced database; information extraction from contract documents using **NLP** and **hierarchical RAG**; feature selection and engineering.
- Delivered the decision-support application (Streamlit) and deployed the infrastructure on **Azure**: ADLS Gen2 data lake, Functions, App Service, containerization, DevOps. Built automated Excel workbooks with VBA macros for reporting outside the tool.

Vice-President of the School's Computing and History Student Associations | ENSAM — France, 2023-2024

- Designed and programmed an ESP32 device for real-time sales recording at a **3,000-attendee** event: MQTT transmission, centralized inventory and **demand forecasting** to anticipate restocking between points of sale.
- Built an annotated virtual tour website; awarded a **€1,000 FSDIE grant**.

Operations Engineering Intern | Blanc-Breizh — France, Summer 2023

- Industrial laundry (300 clients, 180 t/week): operations research problems, scheduling, optimization, delivery routing

EDUCATION

Research Master's (M.A.Sc.), Department of Mathematics and Industrial Engineering | Polytechnique Montréal — Montreal, 2024-2026

Thesis: *Bid response process: integrating artificial intelligence methods that leverage contractual and competitive data.*
Member of: Jarislowsky/AtkinsRéalis Research Chair and CIRRELT (Interuniversity Research Centre on Enterprise Networks, Logistics and Transportation) – Mitacs Accelerate Program
*Results: **GPA 4.0/4.0** – **Excellence Distinction** –accepted peer-reviewed scientific publication (CENTERIS 2026)*

Engineering degree (Master), Grande École Program | Arts et Métiers ParisTech (ENSAM) — France, 2022-2026

Prestigious French engineering school specializing in mechanical, industrial, and technological engineering.
Semester projects with :
- Safran Aircraft Engine ; Instrumented a forging process, then identified the constitutive law using machine learning; validated by numerical simulations.
- CNES (France National Center for Spatial Studies) ; Developed a tool to compute and execute winding trajectories for a composite rocket nose cone.
*Results: **Top 10%** – Recipient of an international mobility scholarship from the alumni association- double degree with Polytechnique Montreal*

Preparatory Classes for the Grandes Écoles (CPGE), PTSI/PT* | Lycée Livet — France, 2020-2022

Intensive preparation for the top French engineering school entrance exams (Mathematics, Physics, Computer science, Engineering sciences)

SELECTED PROJECTS

MechAgent – Agentic AI for Computer-Aided-Engineering (GitHub) — 2026

- An AI engineering agent that transforms natural-language specifications into optimized and verified mechanical components. Orchestrates parametric CAD generation, finite element analysis, material selection, fatigue assessment and mass optimization, with automated engineering reports and STEP export. Proof of concept for L brackets. *(Python, LangGraph, FEM, CalculiX, CadQuery, Optuna, LLM agents)*

From Neuron to Artificial General Intelligence (AGI) (GitHub) — 2026

- Notebooks that rebuild each core component from scratch before using the reference library, then verify that the implementations match it: autograd, Transformer and GPT, BPE tokenizer, LoRA and DPO fine-tuning. RAG, coding agent and MCP server. (Python, PyTorch, Transformers, *RAG, LoRA, evaluation)*

Robotic Pick-and-Place with Reinforcement Learning and World Models (GitHub) — 2025

- A simulated 6-DOF robotic arm pipeline for vision-based object manipulation, including kinematics, RGB-D perception, reinforcement learning (SAC) and learned dynamics models. Developed model predictive control (MPC) with uncertainty-aware planning and benchmarked controllers under varying object properties, positions and physical conditions. *(Python, MuJoCo, PyTorch, Reinforcement Learning, SAC, MPC, World Models, OpenCV)*

LLM Strategy Agent for a Territory-Control Strategy Game (GitHub) — 2026

- Developed an AI agent powered by a local LLM to play OpenFront against the game's built-in AIs. The agent combines strategic planning, a graph-based representation of the game state and rule-based knowledge retrieval. Implemented a strategist–commander architecture. *(TypeScript, agentic AI, RAG, graphs, LM Studio, testing and evaluation)*

SKILLS

General: Complex problem solving, rigor, autonomy, communication, teamwork

Programming: Python (6 years), SQL (4 years)

Data Science: Machine learning, Deep Learning, NLP, RAG, LLMs, AI agents, time series, statistics

Data & Cloud: Azure (ADLS Gen2, Functions, App Service, containers), Docker, Git, Streamlit

Languages: French (native), English (fluent - C1)

Certifications: IBM Data Science Specialization (Coursera), Stanford Machine Learning Specialization (Coursera)